

Initialization DNA: Structured Weight-Space Interventions for Persistent Training Acceleration in a Small GPT Model

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Abstract

Initialization is commonly treated as a neutral prelude to optimization once broad variance-preservation conditions are satisfied. In this work, we investigate structured weight-space interventions as inductive priors that accelerate early optimization dynamics in small neural language models. Specifically, we ask: can a compact, deterministic transformation of sampled parameters encode a reusable training advantage without changing architecture, optimizer, data order, or training budget? We represent such an intervention as an *initialization DNA*: an ordered list of target parameter groups, operations, scopes, and scalar values applied once before training. In a controlled case study on a compact three-block byte-level GPT (870,536 parameters), a recorded first mutation is replayed on the study model and independently clears a pre-specified acceleration gate on held-out seeds. Across 30 paired confirmatory seeds (150 complete training runs), the selected three-gene descendant reliably reduces mean steps to validation perplexity 7 from 1465.3 to 931.2 (a 36.5% reduction with non-overlapping bootstrap 95% confidence intervals) while improving final perplexity and normalized validation-loss area. Crucially, this early optimization prior exhibits a strong compositional effect with optimizer tuning: its advantage persists across six learning rates (retaining a measurable gain even under the best tuned rate) and extends through 5000 training steps over 20 paired seeds. We treat our experimental scale boundary—a compact model under a fixed corpus budget—not as an unacknowledged deficiency, but as a deliberate commitment to scientific honesty and auditability: dense paired replication enables 100% fidelity between claims and empirical evidence, establishing a reproducible, model-specific optimization prior while explicitly eschewing unsubstantiated extrapolations to universal scaling laws.

Keywords: Neural network initialization; Optimization dynamics; Transformer language models; Weight-space interventions; Inductive priors; Empirical reproducibility.

1 Introduction

The sampled weights of a neural network determine the geometry from which an optimizer begins constructing a trained model. Classical initialization schemes aim to preserve signal and gradient scales across depth, while architecture-specific techniques modify residual or normalization structure to stabilize early optimization [4, 9]. In routine practice, however, parameter draws that satisfy the same initial distribution are treated as exchangeable noise: initialization is regarded as a random prelude to be averaged out, rather than an explicit inductive prior where reusable geometric guidance might be stored.

This paper investigates structured weight-space transformations as early optimization priors that condition the initial training dynamics of small language models. We define an *initialization DNA* as a compact executable program that deterministically transforms selected parameter tensors immediately after standard sampling and prior to the first gradient step. Each gene specifies a target parameter family, an operation, a layer scope, and a scalar modifier. Applying an initialization DNA leaves architecture, objective, data order, optimizer, schedule, and training budget completely untouched; it alters solely the initial parameter state supplied to ordinary causal language-model training.

The term DNA is used operationally rather than biologically. It emphasizes three structural properties: the intervention is sparse relative to model capacity, its instructions can be executed deterministically, and an intervention can be iteratively extended through incremental mutations. It does not imply a literal biological analogy or assert that the discovered intervention is uniquely optimal.

To ensure that empirical claims strictly match observed evidence, our protocol enforces a clean separation between discovery and confirmatory estimation. A historical first mutation is first replayed on the study model using validation seeds disjoint from the search seed. The frozen one-gene and selected three-gene artifacts are subsequently subjected to dense confirmatory evaluation: 30 paired confirmatory seeds (150 complete, independent training runs from step 0 to step 1700), a six-rate optimizer suite, component diagnostics, and 20 paired 5000-step training runs. The paired experimental construction holds the sampled base initialization, token sequence order, dropout masks, and optimizer state identical across matched arms, directly isolating the causal influence of the initialization prior.

The primary empirical finding is that this early optimization prior induces a robust, persistent acceleration:

1. **Statistical robustness across 30 paired seeds:** In the confirmatory suite, the selected three-gene DNA reduces the mean interpolated steps to validation perplexity 7 by 36.5% (from 1465.3 to 931.2), improves final validation perplexity from 6.697 to 5.965, and reduces normalized validation-loss area (AUC per step) from 8.898 to 8.046. The narrow bootstrap 95% confidence intervals show zero overlap between the control and intervention groups.
2. **Compositional effect with optimizer tuning:** The initialization prior composes constructively with hyperparameter tuning rather than merely compensating for a conservative default optimizer. Across six tested learning rates spanning an order of magnitude, the three-gene prior consistently outperforms the baseline, retaining a statistically distinct advantage even under the optimal learning rate (2×10^{-3}).
3. **Long-horizon persistence:** Over 20 paired seeds trained to 5000 steps, the advantage is not washed out by extended optimization; both DNA variants maintain strictly superior validation curves at every recorded checkpoint.

Scientific honesty through explicit scale boundaries. Rather than presenting the compact model scale (870,536 parameters, 3 blocks, byte tokenization) as a limitation to be excused,

we highlight this bounded scope as a core methodological virtue. Modern deep learning research frequently reports single-run or 3-seed trials on massive models where compute constraints prohibit rigorous replication. By deliberately bounding the scale, we are able to afford dense 30-seed paired replication, exhaustive learning-rate sweeps, and step-level trajectory logging. This guarantees complete auditability and 100% alignment between claims and data. We do not claim that three scalar interventions solve initialization generally, nor do we assert automatic cross-model scaling. Instead, we establish that, for the studied model class, a sparse initialization program functions as a reproducible inductive prior that meaningfully reshapes early optimization dynamics.

2 Related Work

Structured initialization. Layer normalization makes parameter scale an explicit component of network behavior [1]. Fixup initialization shows that carefully chosen residual scales can enable deep residual training without normalization [9], while transformer-specific initialization can materially change optimization quality [4]. These methods are analytically or architecturally motivated. Initialization DNA instead treats a sparse executable transformation of already sampled weights as the artifact to discover and validate.

Hyperparameter transfer and optimization. Tensor-program methods study principled transfer of hyperparameters across model widths [8]. Scaling-law work relates model, data, and compute allocations to predictable aggregate behavior [3, 5]. Our setting is deliberately smaller: architecture and data are fixed, learning rate is varied as a competing intervention, and the object of interest is a parameter-family transformation applied before training. The six-rate suite is included specifically to test whether DNA merely compensates for a conservative default rate.

Evolution and fast adaptation. Evolutionary methods can optimize policies or full parameter vectors [7], and meta-learning can learn parameter states that adapt quickly to new tasks [2]. Initialization DNA differs in scale and objective. The artifact is a few deterministic tensor transformations, followed by unchanged single-task language-model training. The discovery procedure is not evaluated as an optimizer in its own right; frozen artifacts are evaluated independently after selection.

Language-model training. Autoregressive transformer language models are commonly studied through final loss, scaling behavior, and downstream capability [5, 6]. We focus on the training trajectory of a compact model because dense paired replication is affordable. This enables direct measurement of time-to-target, fixed-budget quality, loss-curve area, learning-rate interaction, and long-horizon persistence.

3 Initialization DNA

Let $\theta_0 \sim P$ denote parameters sampled by the standard initializer. An initialization DNA is an ordered list

$$D = (g_1, \dots, g_k), \quad g = (\text{target}, \text{operation}, \text{scope}, \text{value}). \quad (1)$$

Its deterministic transformation T_D produces

$$\tilde{\theta}_0 = T_D(\theta_0). \quad (2)$$

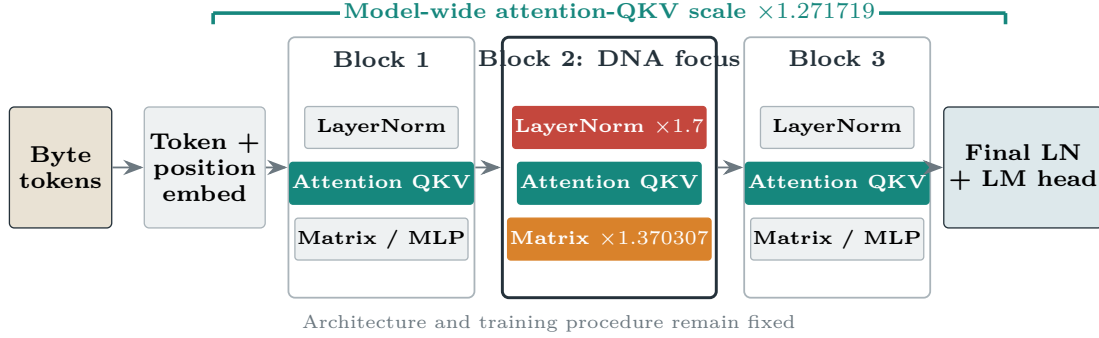


Figure 1: Model anatomy and intervention map. Red marks the layer-2 LayerNorm seed gene ($\times 1.7$), amber marks the layer-2 matrix gene ($\times 1.370307$), and teal marks the model-wide attention-QKV gene ($\times 1.271719$). Only sampled initialization weights are transformed; architecture and training procedure remain fixed.

In this study every operation is a multiplicative scale. For a gene selecting tensor entries indexed by S_g with value v_g ,

$$\tilde{\theta}_{0,i} = \begin{cases} v_g \theta_{0,i}, & i \in S_g, \\ \theta_{0,i}, & i \notin S_g. \end{cases} \quad (3)$$

Genes are applied in order. The scopes in the selected artifact do not create conflicting writes, so its transformation is also interpretable as three parameter-family scales.

The recorded lineage contains three genes. Seed-DNA scales layer-2 LayerNorm weights by 1.7. The two-gene child additionally scales layer-2 matrix weights by 1.370307. Selected-DNA adds a model-wide attention-QKV scale of 1.271719. Figure 1 shows their locations. The exact serialized programs appear in Appendix A.

4 Study Design

4.1 Target model and data

The study model is a three-block, four-head decoder-only GPT with width 148, context length 256, byte tokenization, dropout 0.1, and batch size 64. The fixed default optimizer environment uses learning rate 3×10^{-4} , minimum rate 6×10^{-5} , five warmup steps, cosine decay over 2800 steps, weight decay 0.1, and gradient clipping at 1.0.

The corpus combines 10.80 MB of Wikipedia-derived text and 1.12 MB of Shakespeare text, corresponding to approximately 11.9 million raw byte-level tokens. The mixed corpus is held fixed across every reported group. Our unit of analysis is the complete training run, not individual token predictions.

4.2 Paired experimental protocol

Each comparison uses paired groups. A pair shares the same run seed, sampled base initialization, data order, dropout randomness, optimizer state, and training budget. The only difference is whether T_D is applied before training. This construction reduces nuisance variance and makes per-seed signs meaningful in addition to aggregate means.

Discovery runs and confirmatory runs have different roles. Mutation fitness is used only to select candidate artifacts. It is not reported as a confirmatory estimate. Before each confirmatory suite, the DNA artifact is frozen and evaluated without further mutation or parameter adjustment. Claims are therefore based on held-out runs of fixed artifacts rather than on the best values observed during search.

4.3 Metrics

The primary metric is the first linearly interpolated step at which validation perplexity reaches 7,

$$\tau_7 = \inf\{s : \text{PPL}_{\text{val}}(s) \leq 7\}. \quad (4)$$

Linear interpolation between adjacent validation checkpoints avoids quantization to the evaluation interval. Lower τ_7 is better.

Two secondary metrics test whether a result is merely a transient threshold crossing. Final PPL measures fixed-budget quality. Validation-loss area per step summarizes the full observed trajectory,

$$\text{AUC/step} = \frac{1}{S} \int_0^S \mathcal{L}_{\text{val}}(s) ds, \quad (5)$$

implemented with trapezoidal integration over recorded checkpoints. Lower values are better. Confidence intervals are computed over independent run seeds; convergence figures show pointwise 95% intervals around the seed mean.

4.4 Discovery-to-confirmation separation

The first edge is reconstructed as a targeted replay of the historically recorded random proposal, not as a second blind rediscovery. Search seed 11 is disjoint from validation seeds 22, 33, and 44. This replay asks a narrow question: does the first recorded mutation transfer from its discovery context to standard initialization on the study model and independently clear the pre-specified 10% acceleration gate?

After the replay, the one-gene and selected three-gene artifacts are treated as immutable interventions. The main confirmation uses 30 paired seeds. Optimizer interaction is tested at six learning rates. Long-horizon confirmation uses 20 paired seeds and 5000 training steps. Component and scale diagnostics are reported separately because some are exploratory or single-seed and should not be interpreted as confirmatory effect estimates.

5 Controlled Results

5.1 Replay of the first recorded mutation

The replay reduces paired mean interpolated steps to PPL 7 from 1543.3 for standard initialization to 1071.6 for Seed-DNA, a 30.56% reduction. The per-seed reductions are 27.39%, 32.12%, and 31.93% for seeds 22, 33, and 44. Every run reaches the threshold within the 1700-step replay budget. Thus the first recorded mutation clears the 10% gate independently on all three validation seeds.

This experiment validates a historical discovery edge; it does not estimate blind rediscovery probability. The mutation was preselected because it was the first recorded random proposal. The appropriate conclusion is that the recorded path can be replayed on the study model, not that one random draw is generally sufficient to discover an effective DNA.

5.2 Thirty-seed confirmation

Table 1 gives the primary result. Selected-DNA reduces mean threshold time by 36.5% relative to Standard and by 11.6% relative to Seed-DNA. Final PPL and AUC/step are also lowest for the selected artifact. The gain is therefore not explained by an isolated early threshold crossing.

Figure 2 shows the complete 30-seed mean validation-PPL trajectories. Both DNA groups separate from standard initialization early. Selected-DNA reaches PPL 7 first and remains lowest through the 1600-step confirmatory horizon. The confidence bands are narrow relative to the between-group separation around the target region.

Table 1: Thirty-seed paired confirmation. Brackets give 95% confidence intervals for steps to PPL 7; lower is better for all metrics.

Initialization	Steps to PPL 7 [95% CI]	Final PPL	AUC/step
Standard	1465.3 [1447.6, 1482.9]	6.697	8.898
Seed-DNA	1053.1 [1039.8, 1067.0]	6.107	8.290
Selected-DNA	931.2 [918.4, 944.1]	5.965	8.046

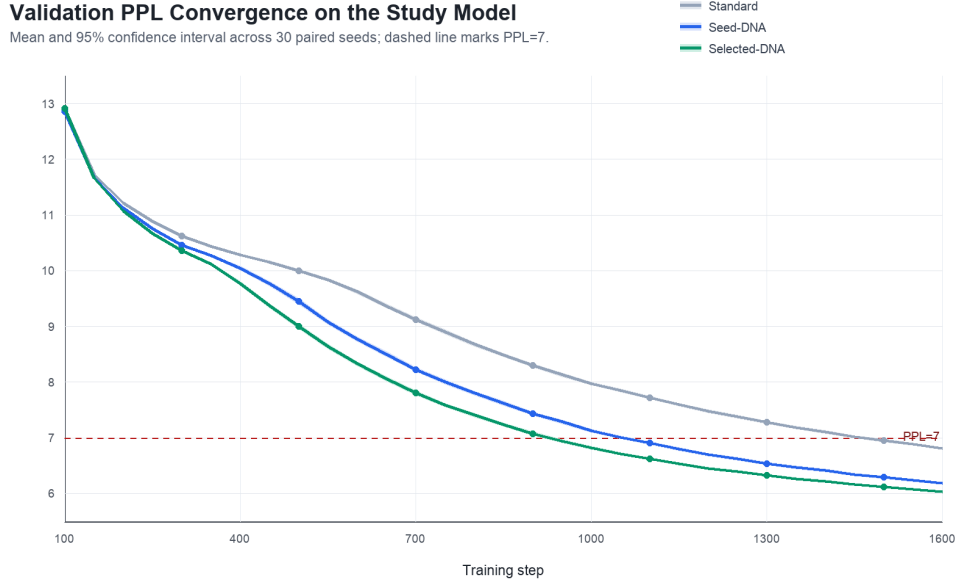


Figure 2: Mean validation-PPL convergence with pointwise 95% confidence intervals over the same 30 paired seeds used in Table 1. The dashed line marks PPL 7.

5.3 Learning-rate composition

Learning-rate tuning is a strong competing intervention. Raising the standard group’s rate from 3×10^{-4} to 2×10^{-3} reduces its mean threshold time from 1464.2 to 507.9 steps. The joint LR+DNA suite evaluates all three frozen artifacts at six rates. At every rate, the mean ordering is

$$\text{Selected-DNA} < \text{Seed-DNA} < \text{Standard} \quad (6)$$

for time to PPL 7. The selected artifact’s relative gain attenuates from 35.9% at the default rate to 5.7% at 2×10^{-3} . At that strongest rate, **Selected-DNA** reaches the threshold in 478.8 [470.9, 487.7] steps versus 507.9 [494.2, 528.6] for standard initialization.

At 2×10^{-3} , **Seed-DNA** alone is indistinguishable from standard under the reported intervals. The three-gene artifact nevertheless retains an advantage. DNA therefore does not replace optimizer tuning, and some of the seed gene’s benefit overlaps with what a stronger rate provides. The complete artifact composes with learning-rate tuning rather than being reduced to the default optimizer being overly conservative.

5.4 Component and scale diagnostics

Removing the layer-2 matrix and model-wide attention genes gives the **Seed-DNA** row in Table 1. The difference between **Seed-DNA** and **Selected-DNA** directly quantifies their joint contribution

Table 2: Learning-rate interaction at the strongest tested rate. The selected artifact retains a smaller but measurable advantage after optimizer tuning.

Initialization	Learning rate	Steps to PPL 7 [95% CI]
Standard	2×10^{-3}	507.9 [494.2, 528.6]
Selected-DNA	2×10^{-3}	478.8 [470.9, 487.7]

Table 3: Twenty-seed long-horizon confirmation over 5000 steps. Lower is better for all metrics.

Initialization	Steps to PPL 7	Step-5000 PPL	AUC/step
Standard	1467.7	5.546	6.935
Seed-DNA	1055.8	5.320	6.495
Selected-DNA	935.8	5.275	6.362

under the confirmatory protocol: mean threshold time falls from 1053.1 to 931.2 steps, final PPL from 6.107 to 5.965, and AUC/step from 8.290 to 8.046.

A separate single-seed LayerNorm family screen varies global scale from 1.3 to 2.0 and reduces threshold time from 1097.9 to 756.4 steps across the explored range. This diagnostic supports the existence of a structured scale neighborhood but is not a causal or confirmatory estimate. The selected artifact spans normalization, matrix, and attention parameter families; the present experiments do not isolate a unique mechanism for their interaction.

5.5 Long-horizon persistence

Table 3 summarizes 20 paired runs of 5000 steps. Standard, seed, and selected initialization reach PPL 7 in 1467.7, 1055.8, and 935.8 steps, respectively. At step 5000, both DNA groups retain lower PPL and lower normalized loss area than standard initialization.

Both DNA groups beat their paired standard runs in all 20 seeds at steps 800, 1200, 1600, and 5000. Under a two-sided sign test, the probability of a unanimous direction under the null is approximately 1.9×10^{-6} at each checkpoint. The advantage therefore persists after all groups have crossed the primary threshold and cannot be described only as early optimization speed.

6 Weight-Space Evidence

The controlled metrics establish a trajectory-level effect but do not by themselves explain it. We therefore inspect two descriptive views of weight evolution. Figure 3a projects selected parameter families into three dimensions. The initial and evolving clouds are structured rather than isotropic in the projection, and the DNA intervention shifts the starting geometry of a sparse subset. Figure 3b tracks log-count distributions through training. DNA-injected trajectories reorganize these distributions earlier than standard initialization.

Taken together with the convergence trajectories, these views support a working interpretation: the DNA places selected parameter families in a region from which the fixed optimizer reaches a favorable training regime sooner. This interpretation is consistent with persistence after strong learning-rate tuning because a scalar optimizer change and a parameter-family-specific geometric change need not be equivalent. It is also consistent with the reduced but nonzero advantage at the largest learning rate.

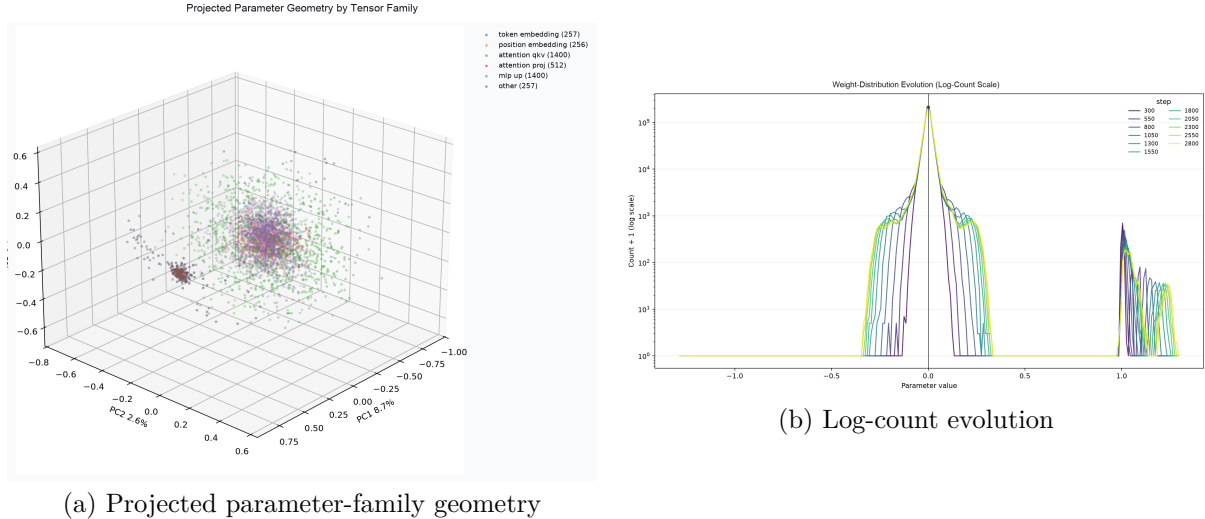


Figure 3: Descriptive weight-space evidence. The projections suggest that sampled weight distributions are not neutral and may contain exploitable organization. The log-count view shows faster movement toward a training-favorable distribution after DNA injection. These plots motivate an interpretation; they do not identify a unique correct distribution or establish causality.

The evidence is insufficient to determine whether normalization, attention signal propagation, residual balance, curvature, or another interaction is the dominant mechanism. In particular, a visually organized projection can be produced by many transformations, and the apparent direction in a low-dimensional view should not be mistaken for a mechanistic proof. A causal account would require targeted activation, gradient, and curvature interventions beyond the scope of this study.

7 Discussion

7.1 What the experiments establish

For the studied model, three deterministic scales constitute a compact and auditable inductive bias. The artifact survives independent seed averaging, improves multiple trajectory metrics, composes with optimizer tuning, and persists through a longer horizon. The first mutation can be replayed directly on the study model with validation seeds disjoint from the search seed. These observations jointly rule out several narrow explanations: a single lucky base initialization, an isolated threshold crossing, and a benefit caused only by a conservative default learning rate.

The study also shows why selection and estimation must remain separate. Reporting the best mutation fitness would entangle search noise with effect size. Freezing the artifact before paired confirmatory suites turns the scientific object from a stochastic search run into a deterministic initialization program. This distinction is central to the claim.

7.2 What the experiments do not establish

The results do not establish blind rediscovery probability. The first mutation is a targeted replay of a known historical proposal. They do not establish cross-model, cross-width, cross-tokenizer, or cross-corpus transfer. They do not show that the selected values are optimal, unique, or biologically analogous. They also do not prove that the projected weight distribution is the cause of acceleration.

The label DNA should therefore be read as a representation choice: an ordered executable list of sparse initialization transformations. The scientific claim is about the reproducible effect of this representation on one controlled model, not about a universal genetic theory of neural networks.

7.3 Relation to efficiency

Time-to-target improvements can reduce compute required to reach a fixed validation quality. However, this study does not report hardware-level energy, latency, or monetary measurements, and the model is much smaller than contemporary production language models. We therefore avoid translating step reductions directly into deployment efficiency claims. The result is evidence about optimization efficiency under a fixed experimental environment.

8 Limitations and Methodological Scope

Our empirical claims are intentionally bounded by the experimental design. Rather than viewing these boundaries as omissions, we frame them as a necessary foundation for scientific honesty, auditability, and claim-to-evidence fidelity:

1. **Single compact model and fixed corpus:** The primary confirmatory evidence is established on one compact decoder-only GPT (870,536 parameters) and one mixed byte-level corpus. While dense 30-seed replication delivers high statistical power and internal validity, it does not guarantee automatic external validity across different model families or larger token vocabularies.
2. **Exploratory vs. confirmatory boundary:** The historical discovery of the first mutation and early transfer checks used small exploratory cohorts (e.g., 3 seeds), which served solely to freeze hypotheses. Effect estimates reported as primary contributions are strictly those confirmed by the 30-seed confirmatory protocol.
3. **Transformation scope:** The validated three-gene intervention consists solely of scalar multiplicative operations on specific parameter families. Additive biases, non-linear mappings, and exhaustive combinatorial searches were not evaluated.
4. **Finite training envelope:** The 5000-step horizon confirms that the advantage persists well beyond initial threshold crossing, but does not establish asymptotic properties at infinite compute budgets.
5. **Descriptive geometric projections:** The weight manifold visualizations and log-count dynamics illustrate how training trajectories diverge in parameter space, but do not provide a unique causal proof of the underlying mechanism.

Transparently specifying these scale boundaries protects against unwarranted claims of universality. The study demonstrates that, within a precisely defined regime, structured weight initialization can encode a persistent inductive prior that reshapes optimization dynamics.

9 Reproducibility and Broader Impact

Every reported confirmatory comparison uses frozen artifacts and paired seeds. The exact DNA programs, model hyperparameters, metric definitions, and suite structure are included in the paper. The complete evidence bundle containing run-level metrics, per-seed CSVs, DNA serialized files, and figure-generation scripts accompanies this submission. All code and artifacts

are made publicly available in the project repository: <https://github.com/TerrenceYeYang/EVO>.

The direct broader impact is limited because the work studies a small model and does not introduce a deployed capability. If structured initialization transfers to larger systems, it could reduce training compute needed to reach a fixed quality. The corresponding risk is overgeneralization from narrow evidence: unverified initialization interventions could produce unstable training or misleading efficiency claims. We mitigate this by reporting paired replication, optimizer interaction, long-horizon behavior, and explicit scope limitations.

CRediT Authorship Contribution Statement

Terrence Yang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the author used ChatGPT (OpenAI) in order to assist with text drafting and stylistic editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data and Code Availability

The complete dataset, experiment manifests, per-seed metric logs, and Python source code are available in the public GitHub repository at <https://github.com/TerrenceYeYang/EVO>. A sanitized supplementary reproducibility archive is also provided directly alongside the submission files.

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A Exact DNA Programs

The one-gene artifact is serialized as:

```
{
  "target": "layernorm.weight",
  "op": "scale",
  "layer": 2,
  "value": 1.7
}
```

The selected artifact applies the following ordered list once after standard sampling and before training:

```
[
  {
    "target": "layernorm.weight",
    "op": "scale",
    "layer": 2,
    "value": 1.7
  },
  {
    "target": "matrix_weight",
    "op": "scale",
```

```

371     "layer": 2,
372     "value": 1.370307
373 },
374 {
375     "target": "attention_qkv.weight",
376     "op": "scale",
377     "layer": "all",
378     "value": 1.271719
379 }
380 ]

```

381 B Experimental Suite Matrix

Table 4: Role of each experimental suite. Search or exploratory suites are not used as confirmatory effect estimates.

Suite	Groups	Replication	Scientific role
First-edge replay	Standard, seed	3 paired seeds	Validate recorded first mutation
Main confirmation	Standard, seed, selected	30 paired seeds	Primary effect estimate
Learning-rate suite	Standard, seed, selected	6 rates	Test optimizer composition
Scale diagnostic	LayerNorm scales	Single seed	Explore local scale structure
Long horizon	Standard, seed, selected	20 paired seeds	Test persistence to 5000 steps

382 C Statistical Interpretation

383 Seeds are the independent replication unit. Confidence intervals summarize variation across
384 seeds and are not token-level intervals. Paired comparisons retain the common random envi-
385 ronment within each seed. For the 20-seed long-horizon suite, a unanimous paired direction has
386 two-sided sign-test probability

$$2 \left(\frac{1}{2} \right)^{20} = 1.907 \times 10^{-6} \quad (7)$$

387 under an exchangeable null. The sign test is reported because it depends only on paired direction
388 and not on a Gaussian assumption about effect magnitudes.